

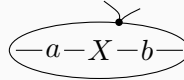
Round 2: every approach targets the book's *current* notation

All approaches now share the book's real `tikz-styles.tex` (`triangle`, `copydot`, `\dloop`, `\dwhiskers`). Test: eq. (70)-style derivative row with the Penrose loop.

A. Legs as node options

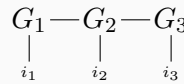
A1: derivative of $a^T X b$ (Penrose loop)

```
\begin{tikzpicture}[tn]
  \node[legs={180}] (a) {$a$};
  \node[right=.7em of a] (X) {$X$};
  \node[right=.7em of X, legs={0}] (b) {
    $b$};
  \draw (a) -- (X) -- (b);
  \dloop[55]{{(a)(b)}}
  \dwhiskers
\end{tikzpicture}
```



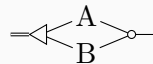
A2: TT chain

```
\begin{tikzpicture}[tn]
  \node[legs={270/i_1}] (G1) {$G_1$};
  \node[right=1em of G1, legs={270/i_2}] (
    G2) {$G_2$};
  \node[right=1em of G2, legs={270/i_3}] (
    G3) {$G_3$};
  \draw (G1) -- (G2) -- (G3);
\end{tikzpicture}
```



A3: Khatri–Rao $A * B$ (book geometry)

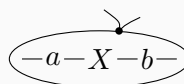
```
\begin{tikzpicture}[tn]
  \node[triangle, rotate=180] (f) {};
  \node (A) at (.6,.25) {A};
  \node (B) at (.6,-.25) {B};
  \node[copydot, legs={0}] (c) at (1.2,0)
    {};
  \draw (f) -- (A) -- (c);
  \draw (f) -- (B) -- (c);
  \draw[double] (f) -- ++(-.4,0);
\end{tikzpicture}
```



B. The graphs library

B1: chain + loop drawn after

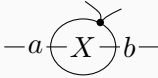
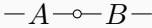
```
\begin{tikzpicture}[tn]
\graph[grow right sep=.7em] {
  l[as=, coordinate] -- a[as=$a$] --
  X[as=$X$] -- b[as=$b$] --
  r[as=, coordinate]
};
\dloop[55]{{(a)(b)}}
\dwhiskers
\end{tikzpicture}
```



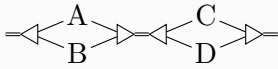
B3: Khatri–Rao fan — does not work. Two attempts (branch down sep, explicit yshifts) could not

make the `graphs` placement engine center a fan-out/fan-in between two triangles; the branch nodes land on or below the chain axis and downstream nodes misalign. Verdict: approach B is pleasant for *chains* (B1 composes fine with `\dloop`), but branching shapes should use approach A or D.

C. Mini-DSL (scope: inline chains)

C1: derivative row — loop via trailing ’	
<code>\tn{-a-{X’}-b-}</code>	
C2: Hadamard-style chain	
<code>\tn{-A-*-B-}</code>	

D. Pics (molecules)

D2: mixed product (511) by composition	
<pre> \begin{tikzpicture}[tn] \pic (k1) {kron={A}{B}}; \pic (k2) at (1.7,0) {kron={C}{D}}; \draw[double] (k1-in) -- ++(-.35,0); \draw[double] (k1-out) -- (k2-in); \draw[double] (k2-out) -- ++(.35,0); \end{tikzpicture} </pre>	
D3: determinant stack	
<pre> \begin{tikzpicture}[tn] \pic {det stack=X}; \end{tikzpicture} </pre>	

E. Paper reproduction test (approach A + `\dloop`)

The forms the book actually uses, including subset loops.

R1: eq (70) full derivation — whole loop, X-only loop, result

```

\begin{tikzpicture}[tn]
  \node (a) {\$a\$};
  \node[right=.7em of a] (X) {\$X\$};
  \node[right=.7em of X] (b) {\$b\$};
  \draw (a) -- (X) -- (b);
  \dloop[55]{(a)(b)}
  \dwhiskers
\end{tikzpicture}
$=$
\begin{tikzpicture}[tn]
  \node (a) {\$a\$};
  \node[right=1em of a] (X) {\$X\$};
  \node[right=1em of X] (b) {\$b\$};
  \draw (a) -- (X) -- (b);
  \dloop[55]{(X)}
  \dwhiskers
\end{tikzpicture}
$=$
\begin{tikzpicture}[tn]
  \node (a) {\$a\$};
  \node[right=2.2em of a] (b) {\$b\$};
  \draw (a.east) -- ++(.25,.28);
  \draw (b.west) -- ++(-.25,.28);
\end{tikzpicture}

```

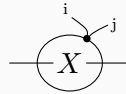
$$a-X-b = a \text{---} X \text{---} b = a \text{---} \text{---} b$$

R2: eq (73) — lone X, labeled whiskers, free edges through the loop

```

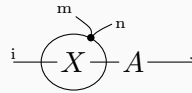
\begin{tikzpicture}[tn]
  \node (X) {\$X\$};
  \draw (X) -- ++(-.8,0);
  \draw (X) -- ++(.8,0);
  \dloop[55]{(X)}
  \draw (ddot) .. controls +(80:.12) ..
    ++(-.2,.28)
    node[pos=1, anchor=south east, font=\tiny] {i};
  \draw (ddot) .. controls +(45:.12) ..
    ++(.28,.18)
    node[pos=1, anchor=west, font=\tiny] {j};
\end{tikzpicture}

```



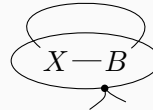
R3: eq (74) — X-only loop inside a longer chain

```
\begin{tikzpicture}[tn]
  \node (X) {$X$};
  \node[right=1em of X] (A) {$A$};
  \draw (X) -- ++(-.8,0)
    node[pos=1, above, font=\tiny] {i};
  \draw (X) -- (A);
  \draw (A) -- ++(.8,0)
    node[pos=1, above, font=\tiny] {j};
  \dloop[55]{(X)}
  \draw (ddot) .. controls +(80:.12) ..
    ++(-.2,.28)
    node[pos=1, anchor=south east, font=\tiny] {m};
  \draw (ddot) .. controls +(45:.12) ..
    ++(.28,.18)
    node[pos=1, anchor=west, font=\tiny] {n};
\end{tikzpicture}
```



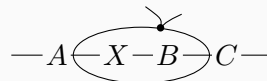
R4: trace row (108) — arc crossing the loop, dot below

```
\begin{tikzpicture}[tn]
  \node (X) {$X$};
  \node[right=1em of X] (B) {$B$};
  \draw (X) -- (B);
  \path (X) edge[out=140, in=40, looseness=2.6] (B);
  \dloop[-55]{(X)(B)}
  \dwhiskersdown
\end{tikzpicture}
```



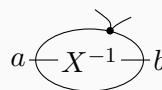
R5: subset loop spanning two nodes mid-chain

```
\begin{tikzpicture}[tn]
  \node (A) {$A$};
  \node[right=1em of A] (X) {$X$};
  \node[right=.7em of X] (B) {$B$};
  \node[right=1em of B] (C) {$C$};
  \draw (A) -- (X) -- (B) -- (C);
  \draw (A) -- ++(-.6,0);
  \draw (C) -- ++(.6,0);
  \dloop[55]{(X)(B)}
  \dwhiskers
\end{tikzpicture}
```



R6: splitting-rule mid-step — loop around an inverse

```
\begin{tikzpicture}[tn]
  \node (a) {$a$};
  \node[right=1em of a] (X) {$X^{-1}$};
  \node[right=1em of X] (b) {$b$};
  \draw (a) -- (X) -- (b);
  \dloop[55]{(X)}
  \dwhiskers
\end{tikzpicture}
```



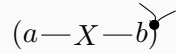
F. Before/after gallery (old code verbatim from the book)

G1. Derivative row (70) LHS

new = if Penrose loops are adopted

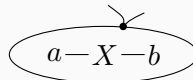
OLD (9 lines)

```
\begin{tikzpicture}[baseline=(a0.base),
  inner sep=1pt]
  \node (a0) {\$(a\$};
  \node[right=1em of a0] (a1) {\$X\$};
  \node[right=1em of a1] (a2) {\$(b)\$};
  \path (a0) edge (a1);
  \path (a1) edge (a2);
  \draw[d] (a2.east) .. controls
    +(80:.12) .. ++(-.2,.28);
  \draw[d] (a2.east) .. controls
    +(45:.12) .. ++(.28,.18);
\end{tikzpicture}
```



NEW (8 lines)

```
\begin{tikzpicture}[tn, baseline=(a0.base)
]
  \node (a0) {\$(a\$};
  \node[right=.8em of a0] (a1) {\$X\$};
  \node[right=.8em of a1] (a2) {\$(b)\$};
  \draw (a0) -- (a1) -- (a2);
  \dloop[55]{(a0)(a2)}
  \dwhiskers
\end{tikzpicture}
```



G2. Trace row (108), middle step

loop + down whiskers

OLD (6 lines)

```
\begin{tikzpicture}[baseline=(a1.base),
  inner sep=1pt]
  \node (a1) {\$(X)\$};
  \path (a1) edge[out=160, in=20, loop,
    looseness=4] (a1);
  \draw[d] (a1.south east) .. controls
    +(280:.12) .. ++(-.2,-.28);
  \draw[d] (a1.south east) .. controls
    +(315:.12) .. ++(.28,-.18);
\end{tikzpicture}
```



NEW (6 lines)

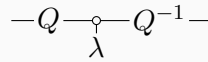
```
\begin{tikzpicture}[tn, baseline=(a1.base)
]
  \node (a1) {\$X\$};
  \path (a1) edge[out=150, in=30, loop,
    looseness=7] (a1);
  \dloop[-55]{(a1)}
  \dwhiskersdown
\end{tikzpicture}
```



G3. Eigendecomposition (1.1)

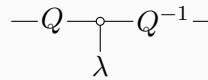
OLD (11 lines)

```
\begin{tikzpicture}[baseline=(Q.base),
  inner sep=1pt]
\node (Q) {$Q$};
\draw (Q.west) -- ++(-.3, 0);
\node[right=1em of Q, copydot] (dot) {};
\path (Q) edge (dot);
\node[below=.3em of dot] (e) {$\lambda$};
\path (dot) edge (e);
\node[right=1em of dot] (Qi) {$Q^{-1}$};
\path (dot) edge (Qi);
\draw (Qi.east) -- ++(.3, 0);
\end{tikzpicture}
```



NEW (7 lines)

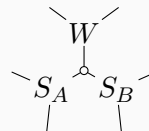
```
\begin{tikzpicture}[tn, baseline=(Q.base),
  every leg label/.style={}]
\node[legs={180}] (Q) {$Q$};
\node[right=1em of Q, copydot,
  legs={270/\lambda}] (dot) {};
\node[right=1em of dot, legs={0}] (Qi) {
  $Q^{-1}$};
\draw (Q) -- (dot) -- (Qi);
\end{tikzpicture}
```



G4. Strassen star

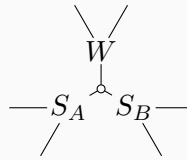
OLD (19 lines)

```
\begin{tikzpicture}[inner sep=1pt,
  baseline=.0em]
\node[copydot] (m) at (0,0) {};
\node (W) at (0,.5) {$W$};
\node (SB) at (-30:.5) {$S_B$};
\node (SA) at (-150:.5) {$S_A$};
\node (w1) at (60:1) {};
\node (w2) at (120:1) {};
\node (sa1) at (-120:1) {};
\node (sa2) at (-180:1) {};
\node (sb1) at (-0:1) {};
\node (sb2) at (-60:1) {};
\draw (m) -- (W);
\draw (m) -- (SA);
\draw (m) -- (SB);
\draw (W) -- (w1) {};
\draw (W) -- (w2) {};
\draw (SB) -- (sb1) {};
\draw (SB) -- (sb2) {};
\draw (SA) -- (sa1) {};
\draw (SA) -- (sa2) {};
\end{tikzpicture}
```



NEW (7 lines)

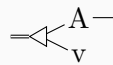
```
\begin{tikzpicture}[tn, baseline=0em, leg
length=.5]
\node[copydot] (m) at (0,0) {};
\node[legs={60,120}] (W) at (0,.5)
{$W$};
\node[legs={-120,180}] (SA) at (-150:.5)
{$S_A$};
\node[legs={0,-60}] (SB) at (-30:.5)
{$S_B$};
\draw (m) -- (W) (m) -- (SA) (m) -- (
SB);
\end{tikzpicture}
```



G5. Kronecker-vector product $A \otimes v$

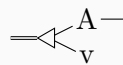
OLD (9 lines)

```
\begin{tikzpicture}[baseline=-.25em, inner
sep=1pt]
\node[triangle, rotate=180] (F0) at
(0,0) {};
\node (A) at (.5,.25) {A};
\node (B) at (.5,-.25) {v};
\draw[double] (F0) -- ++(-.4,0);
\draw (F0) -- (A) -- ++(.5, 0);
\draw (F0) -- (B);
\end{tikzpicture}
```



NEW (6 lines)

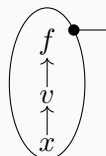
```
\begin{tikzpicture}[tn]
\node[triangle, rotate=180,
double legs={180}] (F0) {};
\node[legs={0}] (A) at (.5,.25) {A};
\node (v) at (.5,-.25) {v};
\draw (F0) -- (A) (F0) -- (v);
\end{tikzpicture}
```



G6. Chain rule (functions ch.) — manual ellipse vs auto-fit

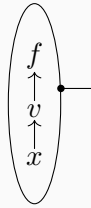
OLD (10 lines; hand-tuned center and radii)

```
\begin{tikzpicture}[baseline=-1em, inner
sep=1pt]
\node (n0) {$f$};
\node[below=1em of n0] (n1) {$v$};
\node[below=1em of n1] (n2) {$x$};
\draw[->] (n2) -- (n1);
\draw[->] (n1) -- (n0);
\drawellipse{0}{-.55}{.5}{1}{180/4}
\end{tikzpicture}
```



NEW (8 lines; ellipse fits itself)

```
\begin{tikzpicture}[tn, baseline=-1em]
  \node (n0) {$f$};
  \node[below=1em of n0] (n1) {$v$};
  \node[below=1em of n1] (n2) {$x$};
  \draw[->] (n2) -- (n1);
  \draw[->] (n1) -- (n0);
  \dloop[30]{(n0)(n1)(n2)}
  \draw (ddot) -- ++(.45,0);
\end{tikzpicture}
```



G7. Order-4 tensor with labeled legs

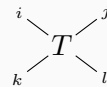
OLD style (9 lines)

```
\begin{tikzpicture}[baseline=-.25em, inner
  sep=1pt]
  \node (T) {$T$};
  \draw (T) -- ++(-.4,.3)
    node[pos=1.3, font=\tiny] {i};
  \draw (T) -- ++(.4,.3)
    node[pos=1.3, font=\tiny] {j};
  \draw (T) -- ++(-.4,-.3)
    node[pos=1.3, font=\tiny] {k};
  \draw (T) -- ++(.4,-.3)
    node[pos=1.3, font=\tiny] {l};
\end{tikzpicture}
```



NEW (3 lines)

```
\begin{tikzpicture}[tn]
  \node[legs={143/i, 37/j, 217/k, -37/l}]
    (T) {$T$};
\end{tikzpicture}
```



G8. Where the new approach does NOT win

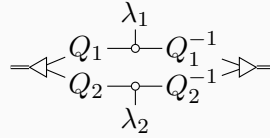
Kronecker sandwich (511): approach A saves only the two outer stub lines; the real compression is approach D's kron pic ($16 \rightarrow 7$, shown in D2 above). **Bent identity wires (511b), twists (510), wrapped traces:** plain TikZ is already minimal — a library adds nothing. **\vecmatvec / \trace / \detstack rows:** the existing macros already beat everything; they stay.

H. Advanced combinations

H1. $\text{eig}(A \otimes B)$, eq (519) — triangles + copydots + labels + bundles

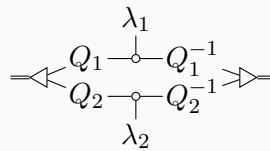
OLD (17 lines)

```
\begin{tikzpicture}[baseline=-.25em, inner
  sep=1pt]
  \node[triangle, rotate=180] (F0) at
    (0,0) {};
  \node (Q1) at (.6,.25) {$Q_1$};
  \node (Q2) at (.6,-.25) {$Q_2$};
  \node[copydot] (b1) at (1.25,.25) {};
  \node[copydot] (b2) at (1.25,-.25) {};
  \node (Q1i) at (2,.25) {$Q_1^{-1}$};
  \node (Q2i) at (2,-.25) {$Q_2^{-1}$};
  \node[triangle] (F1) at (2.7,0) {};
  \draw[double] (F0) -- ++(-.4,0);
  \draw[double] (F1) -- ++(.4,0);
  \draw (F0) -- (Q1) -- (b1) -- (Q1i) --
    (F1);
  \draw (F0) -- (Q2) -- (b2) -- (Q2i) --
    (F1);
  \draw (b1) -- ++(0,.25) node[anchor=
    south] {$\lambda_1$};
  \draw (b2) -- ++(0,-.25) node[anchor=
    north] {$\lambda_2$};
\end{tikzpicture}
```



NEW (12 lines)

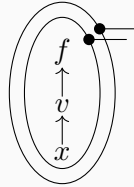
```
\begin{tikzpicture}[tn, leg length=.25,
  every leg label/.style={}]
  \node[triangle, rotate=180,
    double legs={180}] (F0) {};
  \node (Q1) at (.6,.25) {$Q_1$};
  \node (Q2) at (.6,-.25) {$Q_2$};
  \node[copydot, legs={90/\lambda_1}] (b1)
    at (1.25,.25) {};
  \node[copydot, legs={270/\lambda_2}] (b2)
    at (1.25,-.25) {};
  \node (Q1i) at (2,.25) {$Q_1^{-1}$};
  \node (Q2i) at (2,-.25) {$Q_2^{-1}$};
  \node[triangle, double legs={0}] (F1) at
    (2.7,0) {};
  \draw (F0) -- (Q1) -- (b1) -- (Q1i) -- (
    F1);
  \draw (F0) -- (Q2) -- (b2) -- (Q2i) -- (
    F1);
\end{tikzpicture}
```



H2. Second derivative (functions ch.) — NESTED loops

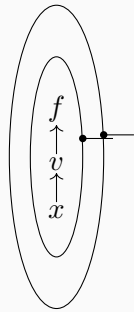
OLD (9 lines; two hand-tuned ellipses)

```
\begin{tikzpicture}[baseline=-1em, inner
  sep=1pt]
  \node (n0) {$f$};
  \node[below=1em of n0] (n1) {$v$};
  \node[below=1em of n1] (n2) {$x$};
  \draw[->] (n2) -- (n1);
  \draw[->] (n1) -- (n0);
  \drawellipse{0}{-.55}{.5}{1}{180/4}
  \drawellipse{0}{-.55}{.7}{1.2}{180/4}
\end{tikzpicture}
```



NEW (10 lines; loops fit themselves and nest)

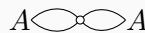
```
\begin{tikzpicture}[tn, baseline=-1em]
  \node (n0) {$f$};
  \node[below=1em of n0] (n1) {$v$};
  \node[below=1em of n1] (n2) {$x$};
  \draw[->] (n2) -- (n1);
  \draw[->] (n1) -- (n0);
  \dloop[35]{(n0)(n1)(n2)}
  \draw (ddot) -- ++(.4,0);
  \dloop[25]{(ddot)}
  \draw (ddot) -- ++(.4,0);
\end{tikzpicture}
```



H3. Wick pairing terms, $\text{Var}[x^T Ax]$ (statistics)

OLD (κ_4 term, 9 lines)

```
\begin{tikzpicture}[baseline=-.25em, inner
  sep=0pt]
  \node (A1) at (0,0) {$A$};
  \node[copydot] (bullet) at (.8,0) {};
  \node[anchor=west] (A2) at (1.4,0) {
    $A$};
  \draw (A1.east) to[out=45,in=135] (
    bullet)
    to[out=45,in=135] (A2.west);
  \draw (A1.east) to[out=-45,in=-135] (
    bullet)
    to[out=-45,in=-135] (A2.west);
\end{tikzpicture}
```

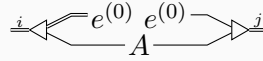


The pairing curves ARE the content here: the library's role is only the `tn` defaults and the `copydot` border behaviour (edges land exactly on the dot). Verdict: parity, no line savings — and none needed.

H4. Direct sum block $e^{(0)} \otimes A \otimes e^{(0)}$ — labeled double legs

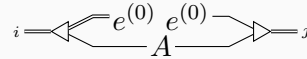
OLD (13 lines)

```
\begin{tikzpicture}[baseline=.25em, inner
  sep=1pt]
  \node (e0l) at (-.35,.4) {\$e^{(0)}\$};
  \node (A) at (0,0) {\$A\$};
  \node (e0r) at (.35,.4) {\$e^{(0)}\$};
  \node[triangle, rotate=180] (F1) at
    (-1.3,.2) {};
  \node[triangle] (F2) at (1.3,.2) {};
  \draw[double] (F1) -- (-.9,.4) -- (e0l)
    ;
  \draw (F1) -- (-.9,0) -- (A);
  \draw (A) -- (.9,0) -- (F2);
  \draw (e0r) -- (.9,.4) -- (F2);
  \draw[double] (F1) -- ++(-0.4,0)
    node[midway, above, font=\tiny] {\$i\$}
    ;
  \draw[double] (F2) -- ++(0.4,0)
    node[midway, above, font=\tiny] {\$j\$}
    ;
\end{tikzpicture}
```



NEW (11 lines)

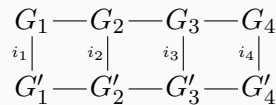
```
\begin{tikzpicture}[tn, baseline=.25em]
  \node (e0l) at (-.35,.4) {\$e^{(0)}\$};
  \node (A) at (0,0) {\$A\$};
  \node (e0r) at (.35,.4) {\$e^{(0)}\$};
  \node[triangle, rotate=180,
    double legs={180/i}] (F1) at
    (-1.3,.2) {};
  \node[triangle, double legs={0/j}] (F2)
    at (1.3,.2) {};
  \draw[double] (F1) -- (-.9,.4) -- (e0l)
    ;
  \draw (F1) -- (-.9,0) -- (A);
  \draw (A) -- (.9,0) -- (F2);
  \draw (e0r) -- (.9,.4) -- (F2);
\end{tikzpicture}
```



H5. TT inner-product zipper — plain TikZ foreach already wins

NEW-style (8 lines; the library only supplies tn)

```
\begin{tikzpicture}[tn]
  \foreach \i in {1,...,4} {
    \node (G\i) at (\i,.45) {\$G_{\i}\$};
    \node (H\i) at (\i,-.45) {\$G'_{\i}\$};
    \draw (G\i) -- (H\i)
      node[midway, left, font=\tiny] {\$i_{\i}\$};
  }
  \foreach \i in {2,...,4} {
    \draw (G\the\numexpr\i-1) -- (G\i);
    \draw (H\the\numexpr\i-1) -- (H\i);
  }
\end{tikzpicture}
```



Regular grids are `foreach` territory (as the book's FFT diagrams already do); a diagram library adds only its

defaults here.